EXPLORATORY DATA ANALYSIS AND VISUALIZATION

Case Study: Wisconsin Diagnostic Breast Cancer Dataset

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Student Name: Hariharan J

Roll Number: _____

Department: Artificial Intelligence and Data Science

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KPR Institute of Engineering and Technology

Abstract

This case study explores the Wisconsin Diagnostic Breast Cancer dataset, containing 569 patient-sample records and 30 numeric measurement features. The objective is to understand feature distributions and relationships, clean and prepare the data, and evaluate a neural-network classifier for benign versus malignant labels. Exploratory analysis includes class distribution, a histogram, a boxplot, a correlation heatmap, and a bivariate scatterplot. Data are divided into stratified training, validation, and test sets; numerical features are standardized using training-set statistics. Three Multi-Layer Perceptron (MLP) configurations are compared on validation accuracy, and the selected model is evaluated on held-out test data. In this fixed-seed run, test accuracy was 0.942 and ROC-AUC was 0.992. These findings demonstrate how visualization and statistical exploration support machine-learning workflows. The model is educational only and must not be used for clinical diagnosis.

1. Introduction and Objectives

Exploratory Data Analysis (EDA) uses summaries and visual representations to understand data quality, distributions, unusual values, and relationships before modeling. This project studies breast-tumor measurement features and uses an MLP neural network as a classification baseline. The practical motivation is to demonstrate data-driven classification and careful evaluation; it is not intended to replace medical professionals.

Objectives

• Inspect the dataset structure, target balance, missing values, duplicates, and summary statistics.

• Perform univariate, bivariate, and multivariate visual analysis.

• Prepare numerical features and create stratified train, validation, and test partitions.

• Compare basic MLP architectures and evaluate the selected model with appropriate metrics and charts.

• Discuss findings, limitations, and future improvements.

2. Dataset Description

The dataset is the Wisconsin Diagnostic Breast Cancer (WDBC) dataset. It contains 569 observations, 30 numeric predictors, a numeric target field, and a readable diagnosis label. Features describe cell-nuclei measurements such as radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, and fractal dimension, including mean, standard-error, and worst-value measurements. Target encoding used here is 0 = malignant and 1 = benign.

Dataset shape: 569 rows × 32 columns. Missing values: 0. Exact duplicate rows: 0. Diagnosis counts: malignant = 212, benign = 357.

Source: UCI Machine Learning Repository, Breast Cancer Wisconsin (Diagnostic). The CSV included with this project allows the analysis to run without a separate download.

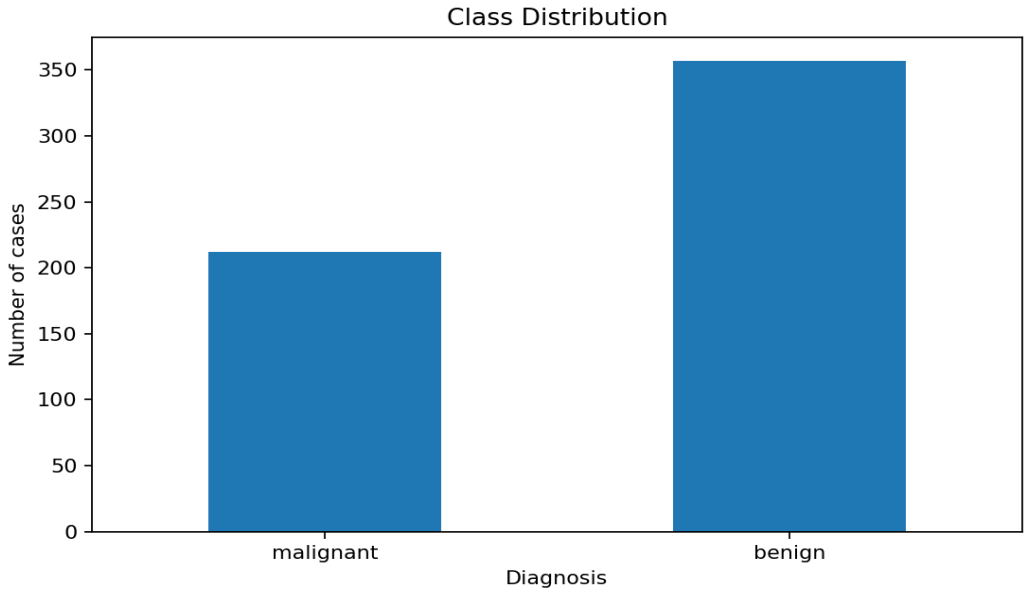
3. EDA and Preprocessing

The analysis checks data types, summary statistics, missing values, duplicates, class balance, feature distributions, and feature correlations. The bundled dataset contains no missing values in the inspected fields. The pipeline includes median imputation as a safeguard for future edits and does not remove outliers automatically: extreme values may be meaningful measurements, so boxplots are used to inspect them rather than deleting them without domain justification. Numerical features are standardized with StandardScaler fitted only on the training partition to prevent information leakage.

A stratified split creates 398 training rows, 85 validation rows, and 86 test rows (approximately 70% / 15% / 15%). Stratification maintains a similar target ratio across the partitions.

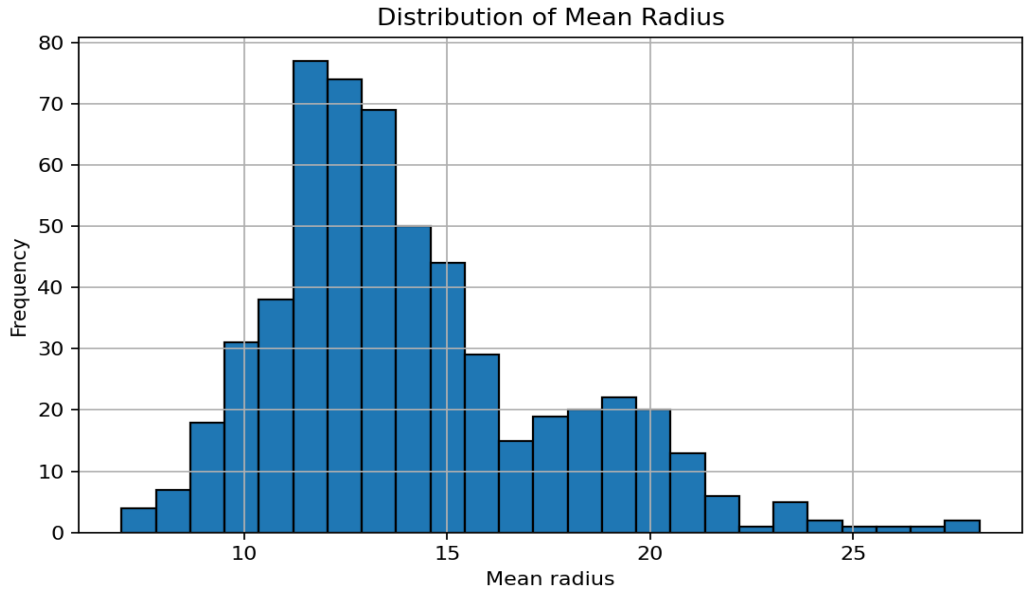
4. Data Visualizations

Figure 1. Class distribution



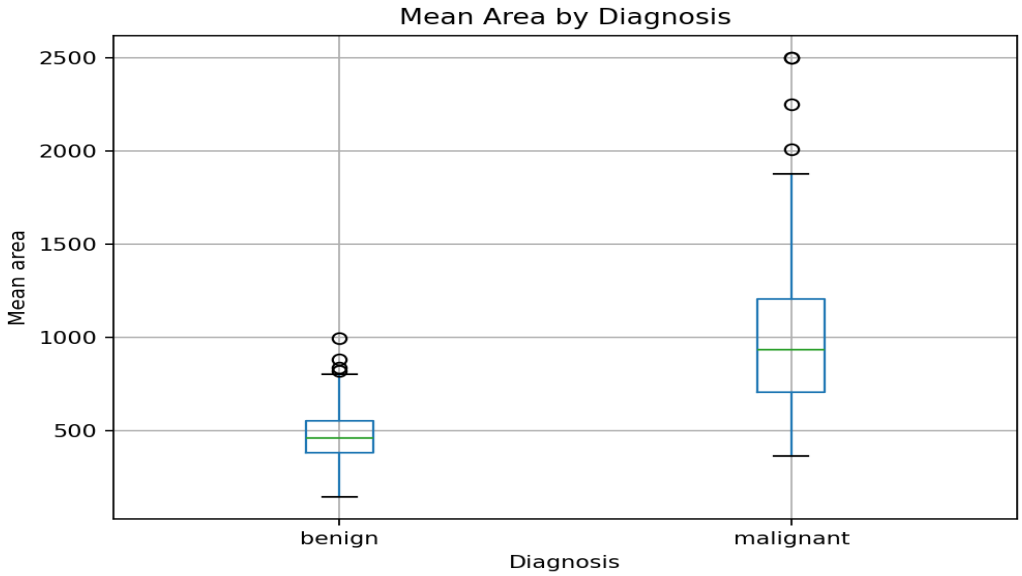
Shows the number of benign and malignant samples. It checks target balance and helps determine whether accuracy should be interpreted alongside class-specific metrics.

Figure 2. Mean-radius histogram



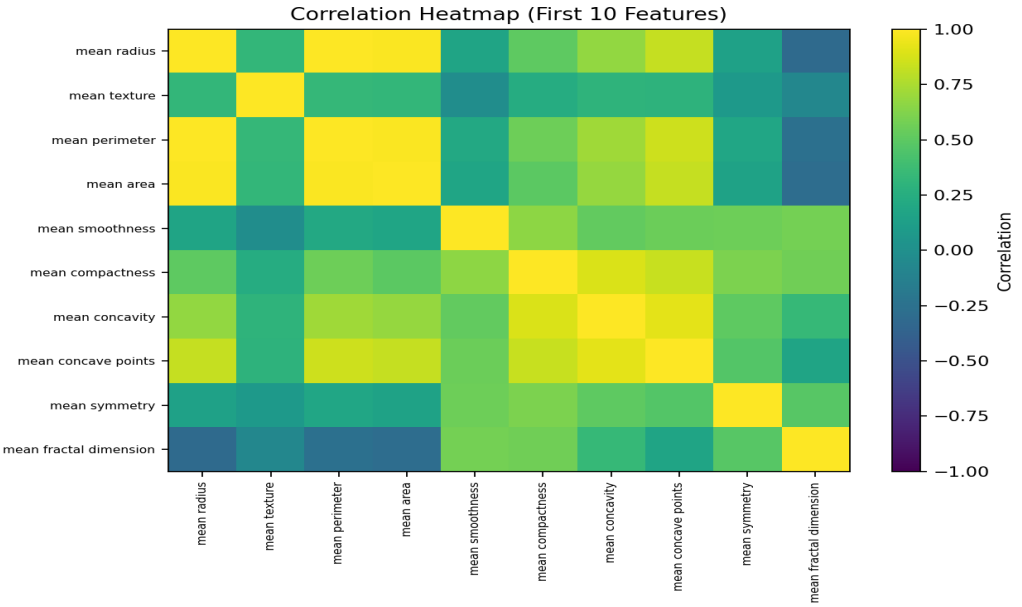
Shows the distribution and spread of mean radius. Skewness or multiple concentrations can indicate that a mean alone is not sufficient to describe a feature.

Figure 3. Mean area by diagnosis



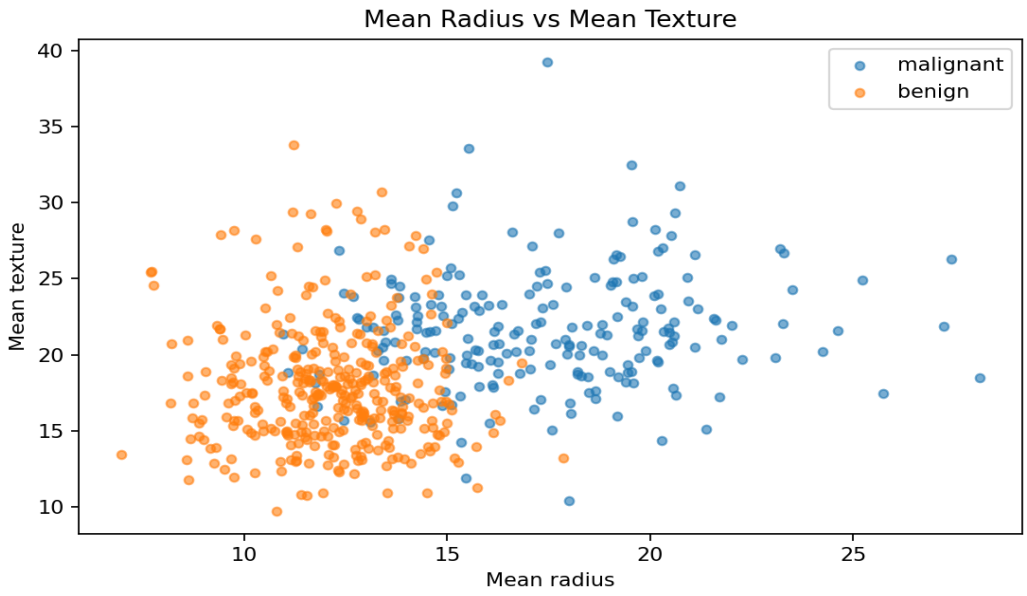
Compares median, spread, and potential outliers of mean area for the two diagnosis groups. It helps assess whether the feature differs across labels.

Figure 4. Correlation heatmap



Displays pairwise linear correlations among the first ten numerical features. Strong correlations indicate related measurements and possible redundancy.

Figure 5. Mean radius versus mean texture



Shows the joint relationship of two measurements and class-wise clustering or overlap. Overlap indicates that a single pair of variables cannot perfectly separate the classes.

Seven-stage visualization process

1. Acquire data and establish the real-world context.
2. Define the analysis question and audience.
3. Select relevant variables and identify their data types.
4. Clean and transform data for reliable analysis.
5. Select visual encodings suited to each question.
6. Add readable titles, axes, scales, legends, and check chart integrity.
7. Interpret the patterns, communicate conclusions, and state limitations.

5. Deep Learning Model

An MLPClassifier neural network from scikit-learn is used as a compact multilayer perceptron model. Input features are standardized. The network uses ReLU activation, Adam optimization, an initial learning rate of 0.001, L2 regularization alpha of 0.001, and early stopping. Three hidden-layer candidates—(32,), (64, 32), and (64, 32, 16)—are compared using validation accuracy. The candidate with the highest validation accuracy is selected before evaluation on the untouched test partition.

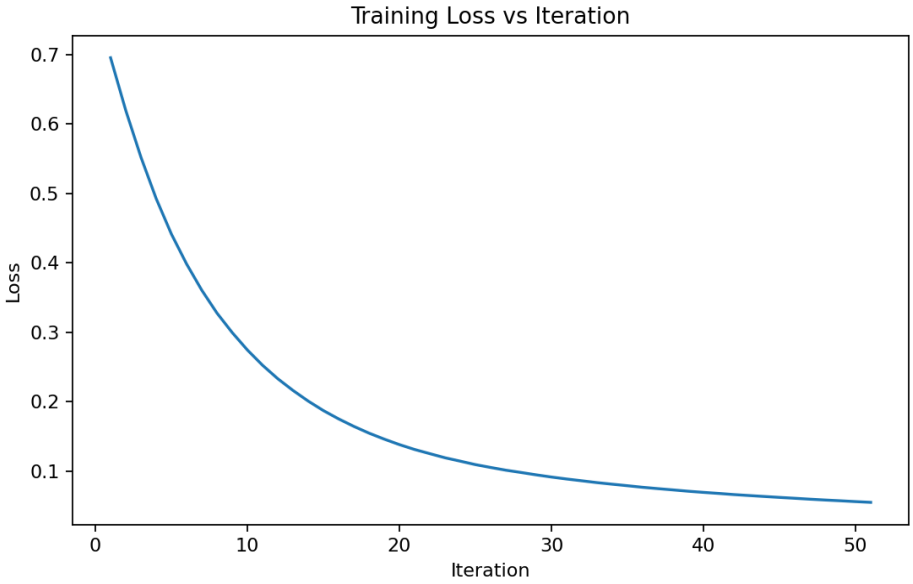
The workflow includes feature analysis through distribution plots and correlation analysis. A permutation-importance chart is included in the runnable script and can be generated when the script is executed. Hyperparameter tuning is basic and limited to hidden-layer configuration; broader tuning is a future improvement.

6. Model Evaluation and Interpretation

Metric	Test result
Accuracy	0.9419
Precision (benign)	0.9298
Recall (benign)	0.9815
F1 score (benign)	0.9550
ROC-AUC (benign)	0.9919
Selected hidden layers	(64, 32)

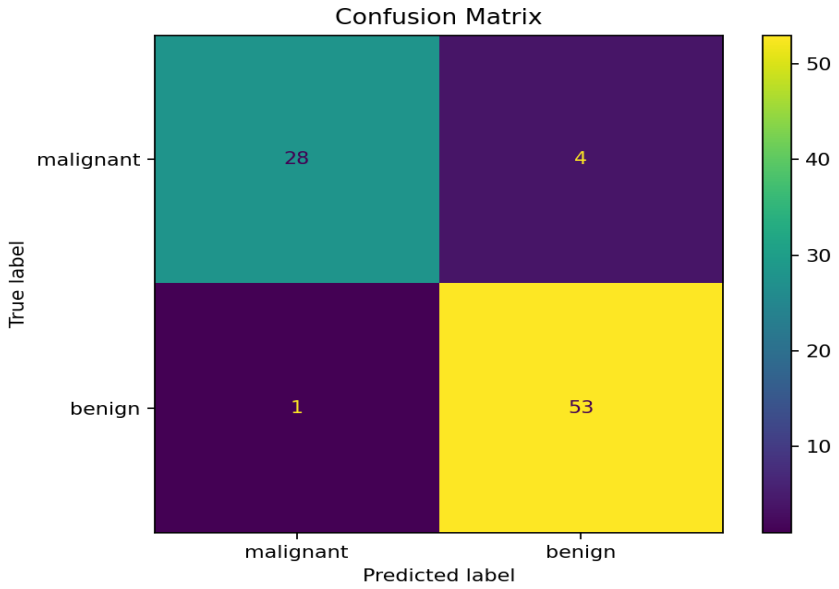
Accuracy is the fraction of correctly classified samples. Precision and recall reported here treat benign as the positive class, so they should be read together with the malignant-class results in the classification report produced by the script. ROC-AUC summarizes ranking performance across probability thresholds. A high score on this small held-out test set does not establish clinical safety or generalization to other hospitals or populations.

Figure 6. Training loss vs iteration



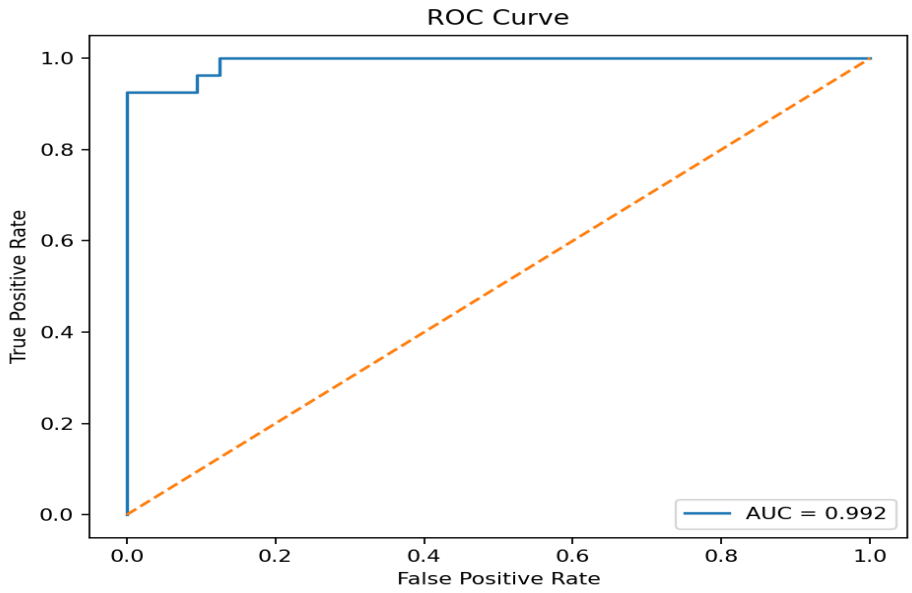
Shows how the optimization loss changes during training. A decreasing curve suggests learning; a plateau can indicate convergence.

Figure 7. Confusion matrix



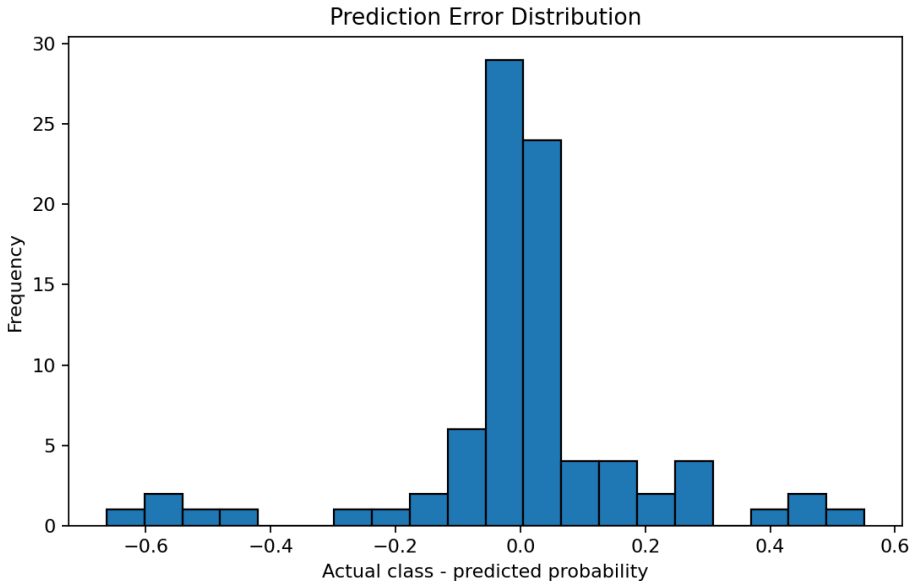
Shows correct and incorrect predictions for both classes, making false positives and false negatives visible.

Figure 8. ROC curve



Plots true-positive rate against false-positive rate across thresholds; the legend reports the area under the curve.

Figure 9. Prediction error distribution



Shows actual encoded class minus predicted benign probability. It gives a rough view of prediction deviations, although class-specific metrics are more directly interpretable.

7. Conclusion and Future Scope

The case study demonstrates a complete EDA workflow on a real-world dataset with more than 15 numerical features. Class distribution, feature distributions, group-wise spread, correlations, and bivariate relationships provide complementary views of the data. The selected MLP model achieved 0.942 test accuracy and 0.992 ROC-AUC in the fixed-seed experiment. These results are promising for a classroom exercise but are not a clinical validation.

Challenges and limitations: The dataset is relatively small, measurements may be correlated, the model is sensitive to the split and software version, and no external validation data were used. Outliers were inspected rather than removed without justification.

Future scope: Perform repeated stratified cross-validation, compare MLP with logistic regression and tree-based models, conduct more systematic hyperparameter search, add calibration and threshold analysis, examine class-specific error costs, and validate on an independent dataset with appropriate clinical oversight.

8. Literature Review and References

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Appendix A: Code and Repository Contents

The full source code is provided in eda_visualization_project.py. The dataset is stored in data/breast_cancer_wisconsin.csv, chart outputs are in visualizations/, and dependencies are listed in requirements.txt. Run the script to reproduce EDA charts, train the MLP, evaluate test metrics, and generate model evaluation plots. README.md contains setup instructions and a repository checklist.

Important: Replace the roll number and submission date on the title page before submitting. Confirm your name and course details, then upload this report and the complete project folder to a public GitHub repository.